Sharp stepsize results for reflected gradient

Abstract

For unconstrained reflected gradient, we prove weak convergence for every globally monotone L-Lipschitz map on a real Hilbert space with a zero throughout $0 < \lambda L < 1/\sqrt{3}$, from arbitrary two-point initialization. An energy that is coercive on actual histories removes a boundedness or summability premise from earlier results. We then prove weak convergence to a variational-inequality solution for every bounded orbit of the exact projected recurrence with an affine monotone operator, throughout the same range. Both first and second differences are square-summable, and the original graph, tangent, and natural residuals vanish. On compact convex sets in finite dimensions, convergence is in norm. Its proof combines five projection inequalities, an exact polynomial certificate, a finite-horizon estimate retaining every boundary term, and the standard forward-reflected-backward distance energy. The inherited rotation example attains the endpoint even on the unit disk. Both results concern the specified recurrence; neither enlarges the universal range of a general maximal-monotone backward splitting.

Keywords: reflected gradient; affine variational inequalities; stepsize bounds; Lyapunov functions; residual convergence

2020 Mathematics Subject Classification. Primary 65K15; Secondary 47H05, 47J20.

1 The algorithm and the convergence statement

Let H be a real Hilbert space. Suppose that $F : H \to H$ is monotone and globally L-Lipschitz, where $L > 0$, and that

$$S = \{z \in H : F(z) = 0\} \neq \varnothing. \tag{1}$$

Thus $\langle F(x) - F(y), x - y\rangle \geq 0$ and $\|F(x) - F(y)\| \leq L\|x - y\|$ for all $x, y \in H$. Given arbitrary $x_{-1}, x_0 \in H$, consider

$$u_n = 2x_n - x_{n-1}, \qquad x_{n+1} = x_n - \lambda F(u_n), \qquad n \geq 0, \tag{2}$$

with a constant $\lambda > 0$. No projection or backward resolvent is used.

The relation to other unprojected single-call methods is exact. For $n \geq 1$, the evaluation points satisfy

$$u_{n+1} = u_n - 2\lambda F(u_n) + \lambda F(u_{n-1}). \tag{3}$$

This is the ordinary unconstrained optimistic-gradient, or forward-reflected, recurrence. The general forward-reflected-backward theorem of Malitsky and Tam [1], with zero backward operator, gives the nonlinear Hilbert-space baseline $\lambda L < 1/2$. These unprojected

equivalences do not identify the algorithms when projections or nonzero backward operators are retained.

At the larger bound $\lambda L < 1/\sqrt{3}$, Nguyen, Trinh and Vuong [2, Proposition 2.1 and Theorem 2.2] give a descent energy and a convergence theorem assuming bounded iterates. Shehu [3, Theorems 10.3–10.4 and Remark 10.7] establishes the affine range and records an additional summability condition in the nonlinear case. We use the numbering of the public versions specified in the bibliography. The purpose here is to prove the unconditional nonlinear statement. The constant and its affine rotation obstruction are already known; the missing ingredient is a coercivity argument that uses the actual preceding evaluation.

Put

$$s = \lambda^2 L^2, \qquad \kappa = \frac{1}{1-2s}, \qquad \varepsilon = \frac{1-3s}{1-2s}, \qquad c_s = \frac{s^2(1-3s)}{(1-s)(1-2s)}. \tag{4}$$

When $0 < s < 1/3$, all four quantities are positive. Define

$$d_n = x_n - x_{n-1} \quad (n \geq 0), \tag{5}$$

and, for $z \in S$ and $n \geq 1$, the energy

$$E_n(z) = \|x_n - z\|^2 + \kappa \left((5s-1)\|d_n\|^2 - 4s\langle d_n, d_{n-1}\rangle + s\|d_{n-1}\|^2\right). \tag{6}$$

Its quadratic form need not be nonnegative on arbitrary independent triples. The algorithmic relation between d_n and u_{n-1} is essential.

Theorem 1. *In the stated setting, suppose $0 < \lambda L < 1/\sqrt{3}$. For every $z \in S$ and $n \geq 1$, the actual trajectory satisfies*

$$E_n(z) \geq c_s\|u_{n-1} - z\|^2 \geq 0, \tag{7}$$

$$E_{n+1}(z) \leq E_n(z) - \varepsilon\|d_{n+1}\|^2. \tag{8}$$

Consequently the orbit is bounded and

$$\sum_{n=1}^{\infty}\|d_{n+1}\|^2 \leq \frac{E_1(z)}{\varepsilon}, \qquad \sum_{n=1}^{\infty}\|F(u_n)\|^2 \leq \frac{E_1(z)}{\varepsilon\lambda^2}. \tag{9}$$

The whole sequence (x_n) converges weakly to a point of S. In finite dimensions it converges in norm. No weak sequential continuity of F is required.

The result is sharp as a universal constant-step statement over this class: Section 5 gives the known rotation instance, with equal unit initial points, for which convergence fails at $\lambda L = 1/\sqrt{3}$ and the norm diverges above it.

2 Positivity on actual histories

The recurrence gives, for every $n \geq 1$,

$$d_n = -\lambda F(u_{n-1}), \qquad u_n - u_{n-1} = 2d_n - d_{n-1}. \tag{10}$$

The initial increment $d_0 = x_0 - x_{-1}$ is arbitrary. Neither an evaluation u_{-1} nor a recurrence preceding $n = 0$ is needed.

Lemma 1. *For $0 < s < 1/3$, the energy (6) satisfies (7) along every trajectory, including $n = 1$.*

Proof. Fix $n \geq 1$ and $z \in S$. Abbreviate

$$v = u_{n-1} - z, \qquad a = d_n, \qquad b = d_{n-1}. \tag{11}$$

Then $x_n - z = v + a - b$, and (10) together with $F(z) = 0$ yields

$$\langle a, v\rangle \leq 0, \qquad \|a\|^2 \leq s\|v\|^2. \tag{12}$$

An expansion of (6), using $1 + 2\kappa s = \kappa$, gives

$$E_n(z) = \|v\|^2 + 2\langle v, a - b\rangle + \kappa\left(3s\|a\|^2 - 2\langle a, b\rangle + (1 - s)\|b\|^2\right). \tag{13}$$

Let

$$\gamma_s = \frac{1 - 3s + 3s^2}{(1 - s)(1 - 2s)} > 0. \tag{14}$$

Completing the square in b gives the exact identity

$$\begin{aligned} E_n(z) = c_s\|v\|^2 + \kappa(1 - s)\left\|b - \frac{a + (1 - 2s)v}{1 - s}\right\|^2 \\ - \frac{2s}{1 - s}\langle a, v\rangle + \gamma_s\left(s\|v\|^2 - \|a\|^2\right). \end{aligned} \tag{15}$$

For verification, the square leaves the coefficients $s/(1 - s)$ on $\|v\|^2$ and $-\gamma_s$ on $\|a\|^2$, and $s/(1 - s) - s\gamma_s = c_s$. Also $1 - 3s + 3s^2 = 3(s - 1/2)^2 + 1/4 > 0$. Every term on the right of (15) is nonnegative by (12). This proves the claimed lower bound even when $5s - 1 < 0$.

At $n = 1$, the same argument uses $v = u_0 - z$, $a = d_1 = -\lambda F(u_0)$ and the arbitrary $b = d_0$. Thus no extra initialization condition is hidden in the certificate. ∎

3 The exact dissipation balance

Lemma 2. *For every $n \geq 1$ and $z \in S$,*

$$\begin{aligned} E_n(z) - E_{n+1}(z) = \varepsilon\|d_{n+1}\|^2 + 2\lambda\langle F(u_n), u_n - z\rangle \\ + \kappa\lambda^2\left(L^2\|u_n - u_{n-1}\|^2 - \|F(u_n) - F(u_{n-1})\|^2\right). \end{aligned} \tag{16}$$

In particular (8) holds.

Proof. Write $e = x_n - z$, $a = d_n$, $b = d_{n-1}$ and $c = d_{n+1}$. Expanding the two energies and using (4) gives the algebraic identity

$$E_{n+1}(z) - E_n(z) + \varepsilon\|c\|^2 = 2\langle c, e + a\rangle + \kappa\left(\|c - a\|^2 - s\|2a - b\|^2\right). \tag{17}$$

On the actual trajectory,

$$e + a = u_n - z, \quad c = -\lambda F(u_n), \quad c - a = -\lambda\big(F(u_n) - F(u_{n-1})\big). \tag{18}$$

Together with (10), these formulas turn (17) into (16). Its last two terms are nonnegative by monotonicity against z and Lipschitz continuity, respectively. ∎

4 Boundedness and whole-sequence convergence

Proof of Theorem 1. The two energy inequalities follow from Lemmas 1 and 2. Fix $z \in S$. For every $n \geq 0$, they give

$$\|u_n - z\|^2 \leq \frac{E_{n+1}(z)}{c_s} \leq \frac{E_1(z)}{c_s}. \tag{19}$$

Since $x_n = (u_n + x_{n-1})/2$, including at $n = 0$, induction yields the explicit bound

$$\sup_{n \geq -1} \|x_n - z\| \leq \max\left\{\|x_{-1} - z\|, \sqrt{E_1(z)/c_s}\right\}. \tag{20}$$

This proves boundedness directly from the energy; it is not an additional assumption.

Summing (8) and using the nonnegativity of the terminal energy proves the first inequality in (9). The second follows from $d_{n+1} = -\lambda F(u_n)$. In particular,

$$d_n \to 0, \qquad F(u_n) \to 0, \qquad \|F(x_n)\| \leq \|F(u_n)\| + L\|d_n\| \to 0. \tag{21}$$

For every $z \in S$, the decreasing nonnegative energy $E_n(z)$ has a finite limit. The correction in (6) tends to zero by (21); therefore

$$\lim_{n \to \infty} \|x_n - z\|^2 \quad \text{exists for every } z \in S. \tag{22}$$

Bounded sequences in a Hilbert space have weakly convergent subsequences. Suppose $x_{n_j} \rightharpoonup \bar{x}$. For every fixed $v \in H$, monotonicity gives $\langle x_{n_j} - v, F(x_{n_j}) - F(v)\rangle \geq 0$. The first factor is bounded and $F(x_{n_j}) \to 0$ in norm. Passing to the limit yields

$$\langle \bar{x} - v, F(v)\rangle \leq 0 \qquad (v \in H). \tag{23}$$

Set $v = \bar{x} + th$, with arbitrary $h \in H$ and $t > 0$. By norm continuity of F, letting $t \downarrow 0$ gives $\langle h, F(\bar{x})\rangle \geq 0$ for every h. Thus $F(\bar{x}) = 0$. This argument does not pass F through a weak limit and requires no weak sequential continuity.

To see that the weak cluster point is unique, let $z, z' \in S$ be two such points. By (22), the sequence

$$\|x_n - z\|^2 - \|x_n - z'\|^2 = -2\langle x_n, z - z'\rangle + \|z\|^2 - \|z'\|^2 \tag{24}$$

has a limit. Along subsequences converging weakly to z and z', that limit equals $-\|z-z'\|^2$ and $\|z-z'\|^2$, respectively. Hence $z = z'$. Boundedness and weak sequential compactness now imply that the whole sequence converges weakly to this point: any subsequence violating weak convergence would have a further weakly convergent subsequence with the same limit. In finite dimensions weak and norm convergence coincide. ∎

The proof also covers the zero operator. In that case $x_n = x_0$ for every $n \geq 0$, regardless of x_{-1}, and $x_0 \in S$. The use of E_1 rather than an artificial E_0 is what keeps arbitrary two-point initialization explicit.

The summability statements in (9) are not a pointwise rate for the last residual. The projected result in Section 6 has an affine operator and a bounded-orbit assumption; no nonlinear projected extension, adaptive-step rule, or general backward-operator enlargement is inferred from Theorem 1.

5 The known rotation obstruction

For completeness, we verify sharpness for the specified equal initialization. The planar rotation is the affine obstruction used in [2, 3].

Proposition 5.1. Let $H = \mathbb{R}^2$, $F = LJ$ with $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and $x_{-1} = x_0 = (1, 0)$. The iteration (2) fails to converge when $\lambda L = 1/\sqrt{3}$. For every $\lambda L > 1/\sqrt{3}$, its norm tends to infinity.

Proof. The operator is monotone, L-Lipschitz and has unique zero 0. Identify $\mathbb{R}^2$ with $\mathbb{C}$, put $t = \lambda L$, and let ζ_n denote the complex iterate. Then

$$\zeta_{n+1} = (1 - 2\mathrm{i}t)\zeta_n + \mathrm{i}t\zeta_{n-1}, \qquad \zeta_0 = 1, \quad \zeta_1 = 1 - \mathrm{i}t. \tag{25}$$

For $t \geq 1/\sqrt{3}$, put $D = \sqrt{4t^2 - 1} > 0$. The roots and solution are

$$r_u = \tfrac{1}{2} - \mathrm{i}(t + D/2), \qquad r_s = \tfrac{1}{2} - \mathrm{i}(t - D/2), \tag{26}$$

$$\zeta_n = \left(\tfrac{1}{2} + \frac{\mathrm{i}}{2D}\right) r_u^n + \left(\tfrac{1}{2} - \frac{\mathrm{i}}{2D}\right) r_s^n, \qquad n \geq 0. \tag{27}$$

Both coefficients are nonzero, so the equal starts do excite the critical or unstable mode.

At $t = 1/\sqrt{3}$, one has

$$r_u = \mathrm{e}^{-\mathrm{i}\pi/3}, \qquad r_s = \tfrac{1}{2} - \frac{\mathrm{i}}{2\sqrt{3}}, \qquad |r_s| = 1/\sqrt{3}. \tag{28}$$

The coefficient of r_u^n is $\mathrm{e}^{\mathrm{i}\pi/3}$. Thus the subsequences $n = 6m$ and $n = 6m+1$ tend to $\mathrm{e}^{\mathrm{i}\pi/3}$ and 1, respectively. The orbit does not converge, although its norm tends to one.

For $t > 1/\sqrt{3}$, $t + D/2 > \sqrt{3}/2$, so $|r_u| > 1$. Since $t - D/2 = 1/[4(t + D/2)]$, one has $|r_s| < 1$. The nonzero unstable coefficient in (27) then implies $|\zeta_n| \to \infty$. ∎

Together, Theorem 1 and Proposition 5.1 identify the strict universal constant-step range for (2) over globally monotone Lipschitz maps with a zero. The contribution of the nonlinear proof is the actual-history coercivity certificate (15), which removes a boundedness or summability premise while preserving the known endpoint obstruction.

6 Bounded affine projected trajectories

Let C be a nonempty closed convex subset of a real Hilbert space H, and let

$$\begin{gathered} F(x) = Mx + d, \qquad d \in H, \qquad L = \|M\| > 0, \\ M : H \to H \text{ is bounded, linear and monotone.} \end{gathered} \tag{29}$$

Fix $\lambda > 0$ and consider the exact projected reflected-gradient recurrence

$$y_k = 2x_k - x_{k-1}, \qquad x_{k+1} = P_C\big(x_k - \lambda F(y_k)\big), \qquad k \geq 0, \tag{30}$$

from $x_{-1}, x_0 \in C$. The operator is defined on all of H, since y_k need not belong to C. Each update requires one value of F and one projection. Since F is affine,

$$F(2x_k - x_{k-1}) = 2F(x_k) - F(x_{k-1}). \tag{31}$$

Thus (30) is also projected forward-reflected-backward splitting in this class. Identity (31) does not hold for general nonlinear maps; neither identity eliminates a projection from constrained Popov iteration.

We use the convention

$$N_C(x) = \{a \in H : \langle a, z - x\rangle \leq 0 \text{ for every } z \in C\}, \qquad x \in C.$$

The solution set and the two residuals considered below are

$$S_C = \{x \in C : 0 \in F(x) + N_C(x)\}, \tag{32}$$

$$r_{\tan}(x) = \inf_{a \in N_C(x)} \|F(x) + a\|, \qquad r_\lambda(x) = \lambda^{-1}\|x - P_C(x - \lambda F(x))\|. \tag{33}$$

Computing r_λ separately would require another projection; no such diagnostic computation is included free in the update cost.

Theorem 2. *Suppose* (29) *holds,* $0 < t = \lambda L < 1/\sqrt{3}$*, and the orbit of* (30) *obeys* $\|x_k\| \leq R < \infty$ *for every* $k \geq 0$*. Then*

$$\sum_{k=1}^{\infty} \|x_{k+1} - 2x_k + x_{k-1}\|^2 \leq \frac{2156(2 + 3t)R^2 + 484\lambda R\|d\|}{18(1 - \sqrt{3}t)}. \tag{34}$$

The set S_C is nonempty. The first increments are square-summable, and the whole sequence converges weakly to a point $\bar{x} \in S_C$. Moreover,

$$v_{k+1} = -\frac{x_{k+1} - x_k}{\lambda} + M(x_{k+1} - 2x_k + x_{k-1}) \in F(x_{k+1}) + N_C(x_{k+1}), \qquad v_{k+1} \to 0. \tag{35}$$

Consequently both residuals in (33) *vanish along the iterates. If H is finite-dimensional, the weak convergence is norm convergence. In particular, when C is compact, boundedness is automatic and*

$$x_k \longrightarrow \bar{x} \in S_C \quad \textit{in norm.} \tag{36}$$

On an unbounded constraint set, boundedness remains a hypothesis. No polynomial last-residual rate or nearest-solution identification is asserted. Solution existence follows from the bounded actual trajectory and is not assumed in advance.

For the exact nonlinear projected recurrence, Malitsky [4, Theorem 3.2] proves weak convergence for $\lambda L < \sqrt{2} - 1$. Cevher and Vũ [5, Theorem 2.1] treat general backward operators at that range. Cai and Zheng [6, Theorem 3] prove a last-iterate tangent-residual rate at the same range. In the affine class, Malitsky and Tam [1, Remark 2.1(b) and Corollary 2.6] give the stronger baseline $\lambda L < 1/2$ when $S_C \neq \varnothing$, without assuming boundedness and with square-summable first increments. The additional range of Theorem 2 is therefore $1/2 \leq \lambda L < 1/\sqrt{3}$, under its stated boundedness hypothesis.

The bounded-orbit result of Nguyen, Trinh and Vuong [2, Theorem 2.2] is unprojected; their Remark 2.1 identifies the constrained-analysis obstruction. Shehu [3, Theorem 5.8 and Remark 5.10] treats affine polyhedral problems at a smaller bound and discusses reaching $1/\sqrt{3}$ after finite optimal-face identification. Theorem 2 requires no face-identification or polyhedral assumption but retains the bounded-orbit hypothesis. An inertial improved-stepsize paper [7] is also relevant, but its full theorem was unavailable for this comparison. We make no exhaustive priority claim regarding that extension.

The proof uses positive combinations of projection inequalities and Fourier positivity. Multiplier methods and finite-memory searches are established tools in absolute stability and integral quadratic constraint analysis [8, 9]. The contribution here is the explicit certificate below and its finite-horizon application to (30). To obtain point convergence we use the distance-energy terms of Malitsky and Tam [1, Proposition 2.3 and Lemma 2.4]; the second-difference budget supplies a summable error in that established energy. No general multiplier or distance-energy principle, or new algorithm, is asserted.

7 An exact finite-sequence certificate

Set

$$\mathcal{J} = \{-4, -1, 0, 4, 5\}, \qquad (a_{-4}, a_{-1}, a_0, a_4, a_5) = (13, 4, 48, 11, 22). \tag{37}$$

The weights are positive and satisfy

$$\sum_{j\in\mathcal{J}} a_j = 98, \qquad \sum_{j\in\mathcal{J}} a_j|j-1| = 242. \tag{38}$$

For a finitely supported sequence $X:\mathbb{Z}\to H$, define

$$\mathcal{Q}(X) = \sum_{k\in\mathbb{Z}} \left\langle X_k - X_{k+1} - \lambda M(2X_k - X_{k-1}), \sum_{j\in\mathcal{J}} a_j (X_{k+j} - X_{k+1}) \right\rangle. \tag{39}$$

Lemma 3. *Under* (29)*, for* $0 < t = \lambda L < 1/\sqrt{3}$ *and every finitely supported* X*,*

$$\mathcal{Q}(X) \geq 18(1-\sqrt{3}t) \sum_{k\in\mathbb{Z}} \|X_{k+1} - 2X_k + X_{k-1}\|^2. \tag{40}$$

Proof. Complexify H and M, using an inner product linear in its first argument. For $\xi = u + \mathrm{i}v$,

$$\|\xi\|^2 = \|u\|^2 + \|v\|^2, \qquad M\xi = Mu + \mathrm{i}Mv.$$

Thus the complexified norm of M is still L, and

$$\mathrm{Re}\langle M\xi, \xi\rangle = \langle Mu, u\rangle + \langle Mv, v\rangle \geq 0, \qquad |\langle M\xi, \xi\rangle| \leq L\|\xi\|^2. \tag{41}$$

No self-adjointness, normality, or spectral decomposition of M is used.

For $z = \exp(\mathrm{i}\theta)$, put $c = \cos\theta$ and

$$b(z) = 13z^{-4} + 4z^{-1} + 48 + 11z^4 + 22z^5 - 98z, \tag{42}$$

$$\gamma(z) = \mathrm{Re}\big((1-z)\overline{b(z)}\big), \qquad m(z) = -(2 - z^{-1})\overline{b(z)}, \qquad w(z) = |z - 2 + z^{-1}|^2. \tag{43}$$

Direct Laurent-polynomial expansion gives

$$\begin{aligned} \gamma &= 8(c-1)^2 U(c), & U(c) &= 18c^3 + 38c^2 + 30c + 19, \\ \mathrm{Re}\, m &= 8(c-1)Q_5(c), & Q_5(c) &= 88c^5 + 22c^4 - 158c^3 - 69c^2 + 4c + 8, \\ \mathrm{Im}\, m &= -8(c-1)^2 V(c)\sin\theta, & V(c) &= 88c^3 + 110c^2 + 48c + 29. \end{aligned} \tag{44}$$

Since $w = 4(c-1)^2$, one also has

$$\gamma - 18w = 16(c-1)^2(c+1)(9c^2 + 10c + 5) \geq 0. \tag{45}$$

Here $9c^2 + 10c + 5 = 9(c + 5/9)^2 + 20/9 > 0$. The two further exact identities needed are

$$3\gamma^2 - (\mathrm{Im}\, m)^2 = 64(c-1)^4(2c-1)^2 P_6(c), \tag{46}$$

$$3\gamma^2 - |m|^2 = 64(c-1)^2 D_8(c), \tag{47}$$

where

$$\begin{aligned} P_6(c) &= 1936c^6 + 6776c^5 + 9736c^4 + 8144c^3 + 4637c^2 + 1604c + 242, \\ D_8(c) &= 10124c^8 + 11312c^7 - 13964c^6 - 18384c^5 \\ &\quad - 976c^4 + 1350c^3 - 753c^2 + 88c + 178. \end{aligned} \tag{48}$$

In particular $P_6(c) > 0$ for $c \geq 0$.

We record a rational verification of the remaining interval signs, so that the certificate does not depend on a numerical frequency grid. If $f(c) = \sum_{r=0}^{n} v_r c^r$ and $c = a + (b-a)u$, its Bernstein coefficients on $[a, b]$ are

$$\beta_i = \sum_{\ell=0}^{i} \frac{\binom{i}{\ell}}{\binom{n}{\ell}} (b-a)^\ell \sum_{r=\ell}^{n} v_r \binom{r}{\ell} a^{r-\ell}, \qquad 0 \leq i \leq n. \tag{49}$$

They satisfy

$$f(a + (b-a)u) = \sum_{i=0}^{n} \beta_i \binom{n}{i} u^i (1-u)^{n-i}.$$

The basis functions are nonnegative and sum to one for $0 \leq u \leq 1$. Substitution into (49) gives the following exact minima:

Polynomial	Interval	Degree	$\min_i \beta_i$
D_8	$[-1, -1/2]$	8	10521/448
D_8	$[-1/2, 0]$	8	1365/112
D_8	$[0, 1/3]$	8	482387/6561
$-Q_5$	$[1/3, 1]$	5	863/243

All entries are positive, proving $D_8 > 0$ on $[-1, 1/3]$ and $Q_5 < 0$ on $[1/3, 1]$. The coefficients are specified by an explicit finite rational formula; the accompanying checker expands these identities and verifies every coefficient.

For $-1 \leq c \leq 1/3$, equations (45) and (47) give $|m| \leq \sqrt{3}\gamma$. For $1/3 \leq c \leq 1$, one has $\operatorname{Re} m \geq 0$ and, by (46), $|\operatorname{Im} m| \leq \sqrt{3}\gamma$. In the first case, (41) yields

$$\operatorname{Re}\big(m\langle M\xi, \xi\rangle\big) \geq -|m|\,|\langle M\xi, \xi\rangle| \geq -\sqrt{3}\gamma L\|\xi\|^2.$$

In the second case, the product of the two real parts is nonnegative, so the same lower bound follows from

$$\operatorname{Re}\big(m\langle M\xi, \xi\rangle\big) \geq -|\operatorname{Im} m|\,|\operatorname{Im}\langle M\xi, \xi\rangle| \geq -\sqrt{3}\gamma L\|\xi\|^2.$$

Thus, at every frequency,

$$\gamma\|\xi\|^2 + \lambda \operatorname{Re}\big(m\langle M\xi, \xi\rangle\big) \geq (1 - \sqrt{3}t)\gamma\|\xi\|^2 \geq 18(1 - \sqrt{3}t)w\|\xi\|^2. \tag{50}$$

At $c = 1$ all four quantities b, m, γ, w vanish; no division at this endpoint is used.

For finitely supported X, let $\widehat{X}(\theta) = \sum_k X_k \exp(-\mathrm{i}k\theta)$. The two factors in (39) transform into

$$\big((1-z)I - \lambda(2 - z^{-1})M\big)\widehat{X}, \qquad b(z)\widehat{X}.$$

Parseval's identity therefore expresses $\mathcal{Q}(X)$ as

$$\frac{1}{2\pi}\int_0^{2\pi} \Big[\gamma\|\widehat{X}\|^2 + \lambda \operatorname{Re}\big(m\langle M\widehat{X}, \widehat{X}\rangle\big)\Big]\, d\theta. \tag{51}$$

Applying (50) and Parseval again proves (40). For these finite sequences Parseval follows by integrating finite sums of exponentials; it is valid on any real Hilbert space after complexification. No square-summability of the actual orbit has been presumed. ∎

8 Finite horizons and point convergence

Proof of Theorem 2. Fix an integer $N \geq 9$ and set $X_k = x_k$ for $0 \leq k \leq N$, with $X_k = 0$ otherwise. Define

$$p_k = X_k - X_{k+1} - \lambda M(2X_k - X_{k-1}), \qquad W_k = \sum_{j \in \mathcal{J}} a_j (X_{k+j} - X_{k+1}).$$

The vector p_k vanishes unless $-1 \leq k \leq N+1$. Put

$$I_N = \{4, \ldots, N-5\}.$$

For $k \in I_N$, all coordinates used in p_k, W_k are actual iterates. Projection normality gives

$$x_k - x_{k+1} - \lambda F(y_k) \in N_C(x_{k+1}), \qquad \langle x_k - x_{k+1} - \lambda F(y_k), x_{k+j} - x_{k+1} \rangle \leq 0.$$

Multiplying by the positive weights and using $F(y_k) = My_k + d$, we obtain

$$\langle p_k, W_k \rangle \leq \lambda \langle d, W_k \rangle, \qquad k \in I_N. \tag{52}$$

Every other potentially nonzero summand has its index in

$$E_N = \{-1, 0, 1, 2, 3\} \cup \{N-4, N-3, N-2, N-1, N, N+1\}. \tag{53}$$

There are exactly eleven such indices, since $N \geq 9$. The padded values also have norm at most R, so

$$\|p_k\| \leq (2+3t)R, \qquad \|W_k\| \leq 196R.$$

The boundary summands may have either sign. Retaining their upper bound gives

$$\sum_{k \in E_N} \langle p_k, W_k \rangle \leq 2156(2+3t)R^2. \tag{54}$$

No projection inequality is applied at these artificial boundary indices, and zero need not belong to C.

The affine term in (52) also reduces to endpoints. For each j, the two sums in

$$\sum_{k=4}^{N-5} (x_{k+j} - x_{k+1})$$

cancel except for at most $|j-1|$ terms at each end. This remains valid when the shifted blocks overlap or their length is less than $|j-1|$: the identity is a finite telescoping sum. All remaining indices lie in $[0, N]$. Hence

$$\left\| \sum_{k \in I_N} W_k \right\| \leq 2R \sum_{j \in \mathcal{J}} a_j |j-1| = 484R. \tag{55}$$

Combining (52)–(55) yields

$$\mathcal{Q}(X) \le 2156(2+3t)R^2 + 484\lambda R\|d\|. \tag{56}$$

For $1 \le k \le N-1$, the second difference of X equals that of x. Discarding only the other nonnegative squared terms in Lemma 3, and then letting $N \to \infty$, proves (34). In particular, the argument neither assumes square-summability of the actual orbit nor bounds x_{-1} by R.

Set $D_k = x_k - x_{k-1}$. The proved series gives $D_{k+1} - D_k \to 0$. For each fixed integer $m \ge 1$,

$$D_n = \frac{x_{n+m} - x_n}{m} + \frac{1}{m}\sum_{j=1}^{m}(D_n - D_{n+j}),$$

and therefore

$$\|D_n\| \le \frac{2R}{m} + \frac{m+1}{2}\max_{0\le i<m}\|D_{n+i+1} - D_{n+i}\|. \tag{57}$$

Taking the upper limit as $n \to \infty$, and then $m \to \infty$, proves $D_n \to 0$. This inference uses boundedness, without assuming square-summability of D_n.

The actual projection relation supplies

$$a_{k+1} = -D_{k+1}/\lambda - F(y_k) \in N_C(x_{k+1}).$$

Consequently

$$F(x_{k+1}) + a_{k+1} = -D_{k+1}/\lambda + M(D_{k+1} - D_k) = v_{k+1} \longrightarrow 0,$$

which proves (35). The tangent residual is at most $\|v_{k+1}\|$. Also $x_{k+1} = P_C(x_{k+1} + \lambda a_{k+1})$, so nonexpansiveness gives

$$r_\lambda(x_{k+1}) \le \|F(x_{k+1}) + a_{k+1}\| = \|v_{k+1}\| \to 0. \tag{58}$$

These are residuals of the original affine variational inequality.

We next prove that S_C is nonempty and identify every weak cluster point. Bounded sequences in a Hilbert space have weakly convergent subsequences, and the closed convex set C is weakly closed. Suppose $x_{n_j} \rightharpoonup \bar{x} \in C$. For every fixed $y \in C$, the actual graph certificate and monotonicity give

$$\langle F(y), y - x_{n_j}\rangle \ge \langle F(x_{n_j}), y - x_{n_j}\rangle \ge \langle v_{n_j}, y - x_{n_j}\rangle. \tag{59}$$

The last term tends to zero because $v_{n_j} \to 0$ in norm and the orbit is bounded. The left term has the weak limit $\langle F(y), y - \bar{x}\rangle$. Thus

$$\langle F(y), y - \bar{x}\rangle \ge 0 \qquad (y \in C).$$

For arbitrary $z \in C$, choose $y = \bar{x} + \sigma(z - \bar{x})$, where $0 < \sigma \le 1$. Dividing by σ and letting $\sigma \downarrow 0$, continuity of F gives $\langle F(\bar{x}), z - \bar{x}\rangle \ge 0$. Hence $\bar{x} \in S_C$. This proves existence and

identifies all weak cluster points, without passing a nonlinear map through a weak limit or assuming a solution in advance.

Fix any $z \in S_C$, and put

$$e_k = x_k - z, \qquad B_k = D_{k+1} - D_k, \qquad \mathcal{E}_k(z) = \|e_k\|^2 - 2\lambda\langle MD_k, e_k\rangle, \quad k \geq 1. \tag{60}$$

These distance-energy terms occur in the FRB analysis of [1, Proposition 2.3 and Lemma 2.4]. Here the second-difference estimate supplies the error budget needed at the larger stepsizes. Projection normality and the VI property give

$$\langle D_{k+1} + \lambda F(y_k), e_{k+1}\rangle \leq 0, \qquad \langle F(x_{k+1}), e_{k+1}\rangle \geq 0.$$

For the second inequality, use

$$\langle F(x_{k+1}), e_{k+1}\rangle = \langle Me_{k+1}, e_{k+1}\rangle + \langle F(z), x_{k+1} - z\rangle \geq 0.$$

In particular, $F(z) = Mz + d$ has not been replaced by zero. Since $F(y_k) = F(x_{k+1}) - MB_k$, the squared-norm identity yields

$$\|e_{k+1}\|^2 - \|e_k\|^2 \leq -\|D_{k+1}\|^2 + 2\lambda\langle MB_k, e_{k+1}\rangle. \tag{61}$$

Subtracting the two cross terms in (60), and using $e_{k+1} - e_k = D_{k+1}$, gives

$$\begin{aligned}\mathcal{E}_{k+1}(z) - \mathcal{E}_k(z) &\leq -\|D_{k+1}\|^2 - 2\lambda\langle MD_k, D_{k+1}\rangle \\ &\leq -\|D_{k+1}\|^2 + 2\lambda\langle MB_k, D_{k+1}\rangle \\ &\leq -\tfrac{1}{2}\|D_{k+1}\|^2 + 2t^2\|B_k\|^2. \end{aligned} \tag{62}$$

The second line uses $D_k = D_{k+1} - B_k$ and monotonicity of M. The last line is Cauchy–Schwarz followed by $2tab \leq b^2/2 + 2t^2a^2$. Thus every inequality follows from the actual single projection, with the affine offset retained until it cancels in the difference.

The energy is not presumed nonnegative on arbitrary histories. On the bounded actual trajectory, for $k \geq 1$ one has $\|D_k\| \leq 2R$ and

$$\mathcal{E}_k(z) = \|e_k - \lambda MD_k\|^2 - \lambda^2\|MD_k\|^2 \geq -4t^2R^2. \tag{63}$$

By (34), the error tail in

$$\mathcal{H}_k(z) = \mathcal{E}_k(z) + 2t^2\sum_{j=k}^{\infty}\|B_j\|^2$$

is finite and tends to zero. Equation (62) gives

$$\mathcal{H}_{k+1}(z) \leq \mathcal{H}_k(z) - \tfrac{1}{2}\|D_{k+1}\|^2.$$

The lower bound (63) proves convergence of $\mathcal{H}_k(z)$ and square-summability of D_{k+1}. It also proves convergence of $\mathcal{E}_k(z)$. Since $D_k \to 0$ and e_k is bounded, its cross term tends to zero. Consequently

$$\lim_{k\to\infty}\|x_k - z\|^2 = \lim_{k\to\infty}\mathcal{E}_k(z) \quad \text{exists and is nonnegative for every } z \in S_C. \tag{64}$$

Taking limits in the telescoped energy inequality also gives the explicit bound

$$\sum_{k=1}^{\infty} \|D_{k+1}\|^2 \le 2\mathcal{E}_1(z) + 4t^2 \sum_{k=1}^{\infty} \|B_k\|^2. \tag{65}$$

Its right side is necessarily nonnegative along an actual trajectory. The finite first increment D_1 does not affect square-summability.

For two weak cluster points $z, z' \in S_C$, equation (64) gives a limit to

$$\|x_k - z\|^2 - \|x_k - z'\|^2 = -2\langle x_k, z - z'\rangle + \|z\|^2 - \|z'\|^2.$$

Along the two cluster subsequences that limit equals $-\|z-z'\|^2$ and $\|z-z'\|^2$, respectively. Thus $z = z'$. As in Section 4, boundedness and weak sequential compactness imply weak convergence of the entire sequence to this point. In finite dimensions this is norm convergence, proving (36). Compact C supplies the boundedness hypothesis automatically. ∎

Proposition 8.1. Under (29), for any closed convex C, the iteration (30) has no nonconstant finite periodic orbit when $0 < \lambda L < 1/\sqrt{3}$.

Proof. A finite periodic orbit is bounded. Theorem 2 makes its second differences square-summable, so those periodic differences vanish identically. Its first increments are therefore constant, and periodicity makes them zero. The orbit is constant. This argument needs no assumed existence of a VI solution. It makes no unconditional boundedness claim. ∎

9 The rotation endpoint on the unit disk

The planar rotation used in Section 5 also gives a compact endpoint example for the projected residual conclusions. This is the same inherited obstruction with a different admissible initialization, not a new counterexample mechanism.

Proposition 9.1. Let $H = \mathbb{R}^2$, let C be the closed unit disk, let $F = J$ be the counterclockwise quarter-turn, and let $\lambda = 1/\sqrt{3}$. With

$$x_{-1} = (1/2, \sqrt{3}/2), \qquad x_0 = (1, 0),$$

the projected iteration (30) has the exact period-six orbit

$$x_k = (\cos(k\pi/3), -\sin(k\pi/3)). \tag{66}$$

Its unique VI solution is zero. At every iterate,

$$\operatorname{dist}(x_k, S_C) = 1, \qquad r_{\tan}(x_k) = 1, \qquad r_\lambda(x_k)^2 = 6 - 3\sqrt{3} > 0. \tag{67}$$

Proof. The exact recurrence identity is

$$x_k - \lambda J(2x_k - x_{k-1}) = x_{k+1}.$$

It follows either by substituting $\exp(-\mathrm{i}k\pi/3)$ in (25), or directly by the angle-addition formulas. Since the resulting point belongs to the unit disk, its projection is itself. Both specified initial points are feasible, as required in Theorem 2.

At an interior point of the disk the normal cone is zero, so the only interior solution is zero. At a unit boundary point x, the normal cone is $\{\alpha x : \alpha \geq 0\}$. The relation $Jx + \alpha x = 0$ would give $\alpha = 0$ by taking the scalar product with x, and then $Jx = 0$, a contradiction. This proves uniqueness. Orthogonality gives

$$\inf_{\alpha \geq 0} \|Jx + \alpha x\| = \inf_{\alpha \geq 0} \sqrt{1 + \alpha^2} = 1.$$

Also $\|x - \lambda Jx\| = 2/\sqrt{3}$, whence

$$P_C(x - \lambda Jx) = \frac{\sqrt{3}}{2} x - \frac{1}{2} Jx.$$

Using the normalization $r_\lambda = \lambda^{-1}\|x - P_C(x - \lambda F(x))\|$ in (33), we obtain

$$r_\lambda(x)^2 = 3\left[(1 - \sqrt{3}/2)^2 + 1/4\right] = 6 - 3\sqrt{3}.$$

Thus each endpoint failure asserted in (67) is checked explicitly. ∎

The strict constant is therefore exact for the universal bounded-affine projected convergence statement, including compact convex constraints in finite dimensions. Boundedness is retained on unbounded constraint sets, and no nearest-solution selection is claimed. The result does not concern arbitrary nonlinear projected operators or general backward resolvents.

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